INF10025 Data Management and Analytics

Tutorial 8 – Discussion Questions for Week 8

1. An organisation named GlenCo Solutions has a number of employees, a name and a phone number. Each employee can be identified by his/her employee id.
Many employees purchase computer equipment (such as a laptop or tablet) to use as part of their job. Each piece of equipment has an id (a barcode supplied by the organisation), a description and a date of purchase. Each piece of equipment only ever belongs to a single employee. Some employees (generally those on a higher salary) have more than one piece of equipment.

- **Create an ERD solution based on the above narrative.**
- **Convert the ERD to a relational schema.**

2. A primary school named Eastern Primary School wishes to democratically choose which animal will become the school mascot. The list of alternatives includes Kangaroo, Koala, Tiger, Elephant, Parrot, Penguin and so on.

Each student may vote for **one or more** of the animals to be the mascot. Each student has a student Id and name. Each of the animals has been assigned a numeric id and name (e.g. 7 Tiger).

- **Create an ERD solution based on the above narrative.**
- **Convert the ERD to a relational schema.**

3. **"The Melbourne Sudoku Championships"** are about to take place.
A number of competitors will compete in the championships. Each competitor has unique competitor id and a name. A number of Sudoku Puzzles have been generated for the championships. Each puzzle has a unique number and a Grade (such as Easy, Hard, Diabolical). E.g. Puzzle No: P222 Grade: Hard. During the championships, a puzzle is simultaneously given to a number of competitors. As each competitor completes a puzzle his/her time is recorded.

The organisers of the champion ships want to be able to produce reports such as these:

Id	Name	Puzzle	Grade	Time
101	Peter Jones	P120	Easy	2:41
101	Peter Jones	P178	Medium	4:53
101	Peter Jones	P222	Hard	8:44

Puzzle	Id	Name	Time
P222	507	Nancy Lee	6:21
P222	101	Peter Jones	8:44
P222	324	Sid Davies	8:58

P222	078	Eric Tan	9:27
P222	041	Merv Hunter	9:27
P222	342	Shirlee Brown	15:42

Create an ERD solution based on the above narrative.
Convert the ERD to a relational schema.

4. **Swinny Publishing** encourages staff to write text books for University subjects. Each text book has a title and a unique ISBN code. Each author has a Name and an AuthorId. Each Book written has a **single** Author. Authors often complete two or three books a year.

   Swinny Publishing want to be able to produce reports such as this:

ISBN	Title	AuthorId	Name
1122334455	Intro to Oracle	11	Peter Joshua
3031012022	Intro to VB.Net	45	Regina Lampert
1122446688	Advanced Oracle	11	Peter Joshua

   Draw the ERD for the above situation.
   Show examples of data stored in all the tables generated by your ERD

5. The business narrative for the previous problem has **changed** slightly. Any of the books published by Swinny Publishing could have **multiple authors**.

   Swinny Publishing want to be able to produce reports such as this:

ISBN	Title	AuthorId	Name
1122334455	Intro to Oracle	11	Peter Joshua
3031012022	Intro to VB.Net	45	Regina Lampert
7771117771	Year 1 Statistics	14	Hamilton Bartholomew
7771117771	Year 1 Statistics	36	Leopold W. Gideon
7771117771	Year 1 Statistics	28	Tex Panthollow
6060606060	Network Security	8	Sylvie Gaudet
6060606060	Network Security	3	Herman Scobie

   Draw a new ERD to reflect the changed requirements.
   Show examples of data stored in all the tables generated by your ERD

6. The business narrative for the previous problem has **changed again**.
   The university has **a number of bookshops** (generally one at each campus).
   Each bookshop has a unique ID (such as Hawthorn, Sarawak etc) plus a manager name.

Before the beginning of each semester, each bookshop orders a number of books from Swinny Press.

The system must record:

- the **date** of the order
- the **qty** of books that were ordered
- the **date** of the books were **delivered**

Each order has a unique ID number. **Only <u>one</u> Book may be ordered on an order.**

Swinny Publishing wants to be able to produce reports such as this:

OrderNo	BookShop	Book ISBN	DateOrdered	Qty Ordered	DateDelivered
3001	Hawthorn	1122334455	15 Jan	50	22 Feb
3002	Hawthorn	3031012022	17 Jan	20	
3003	Hawthorn	7771117771	17 Jan	100	15 Feb
2410	Lilydale	6060606060	4 Jan	30	30 Jan
2411	Lilydale	1122334455	20 Jan	150	
4610	Prahran	6060606060	1 Feb	45	

Draw a new ERD to reflect the changed requirements.
Show examples of data stored in all the tables generated by your ERD

7. The business narrative for the previous problem has **changed for the final time.** Each order may now contain **many** requests for many Books

Each bookshop may make an order such as this:

BookShop	DateOrdered	OrderNo
Hawthorn	7 July	6000

ISBN:	Qty
1122334455	60
3031012022	25
7771117771	120

Swinny Publishing wants to be able to produce reports such as this:

OrderNo	BookShop	Book ISBN	DateOrdered	Qty Ordered	DateDelivered
6000	Hawthorn	1122334455	7 July	60	19 July
6000	Hawthorn	3031012022	7 July	25	25 July
6000	Hawthorn	7771117771	7 July	120	

Draw a new ERD to reflect the changed requirements.
Write the Relational Schema for any WEAK Entity
Show examples of data stored in all the tables generated by your ERD